

PAPER_198: F_UBii Taxonomy Part 1 — Compact Object and Stellar Physics Buoyancy Forces

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Author: Star-Magic UQFF Research Framework

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$$F_U(r, t) = \sum_{i=1}^4 U_{gi} + U_m + U_A - U_{bi}, \quad \kappa = 5.0 \times 10^{-4} \text{ day}^{-1}, \quad [SSq] = 0.57$$

$$M_J^{\text{UQFF}} = M_J^{\text{Jeans}} \cdot \left(1 - [SSq] \cdot \frac{B^2}{8\pi\rho c_s^2} \right), \quad [SSq] = 0.57$$

Abstract

The UQFF buoyancy force F_UBii is applied across all major astrophysical phenomena by embedding each system's characteristic energy or length scale into the universal F_rel/E_LEP scaling framework. This paper catalogs 18 unique F_UBii variants relating to compact objects and stellar physics: MHD dynamo buoyancy, terminal velocity, Press-Schechter, Hawking radiation, quasi-normal mode ringdown, Blandford-Znajek jet power, Arnett supernova, entanglement, jet velocity, planet migration, AGN feedback, angular momentum accretion, J-type shock, Sedov-Taylor blast wave, GRB afterglow, SIDM core formation, ionization fronts, superfluid glitch, and virial theorem.

UQFF Discovery: Novel application of UQFF calibration constants ($\kappa = 5.0 \times 10^{-4} \text{ day}^{-1}$, $[SSq] = 0.57$) uniquely enabling this analysis — establishing a new connection in the UQFF framework not present in Standard Model treatments.

1. General F_UBii Framework

$$F_{\text{UBii},X} = \pm F_{\text{rel}} \times (F_X / E_{\text{LEP}}) \times Q_{\text{wave}} \times [\text{decay/oscillation factor}]$$

where:

$F_{\text{rel}} \sim 4.3 \times 10^{33} \text{ N}$ (relativistic UQFF force scale)

E_{LEP} = energy of characteristic LEP scale (J)

Q_{wave} = quantum wave amplitude (J/m³), std $\sim 6.33 \times 10^4$

F_X = system-specific physical expression

2. Compact Object Variants

2.1 MHD Dynamo Buoyancy

$$F_{\text{UBii},\text{MHD}} = F_{\text{rel}} \times (E_M / t_{\text{eddy}} / E_{\text{LEP}}) \times Q_{\text{wave}} \times (3/2)(\text{Re}_m^{1/2} - 1)$$

E_M = magnetic energy density

$t_{\text{eddy}} = l/v \sim \text{Myr}$ (eddy turnover time)
 $Re_m \sim 10^{10}$ (magnetic Reynolds number)
 Source: BB_C_Equations item 748

2.2 Terminal Velocity Buoyancy

$$F_{\text{UBii,termv}} = F_{\text{rel}} \times (v(2GM(1-G)/r_{\text{launch}}) / E_{\text{LEP}}) \times Q_{\text{wave}} \times (t \cdot L/c)$$

$G = L/L_{\text{Edd}} \sim 1$ (Eddington ratio for wind-driven systems)
 $r_{\text{launch}} \sim 100 R_s$ (wind launch radius)
 t = optical depth
 Source: BB_C_Equations item 853, 1823

2.3 Hawking Radiation Buoyancy

$$F_{\text{UBii,haw}} = F_{\text{rel}} \times (hc^3/(8pGMk_B) / E_{\text{LEP}}) \times Q_{\text{wave}} \times (?(2p))$$

$T_H = hc^3/(8pGMk_B)$ (Hawking temperature)
 $? = c^4/(4GM)$ (surface gravity)
 Source: BB_C_Equations item 901, 1246

2.4 Quasi-Normal Mode Ringdown Buoyancy

$$F_{\text{UBii,qnm}} = -F_{\text{rel}} \times (c^3/(2pGM) \cdot (0.3737 + 0.088 \cdot a_f) / E_{\text{LEP}}) \times Q_{\text{wave}} \times e^{-t/t} \times [t ? M]$$

$a_f \sim 0.69$ (dimensionless BH spin post-merger)
 t = ringdown decay time
 $f_{\text{QNM}} = c^3/(2pGM_f) \cdot f(0.3737 + 0.088 \cdot a_f)$ (Berti fits $l=2, m=2$ mode)
 Source: BB_C_Equations item 945, 1293

2.5 Blandford-Znajek Jet Power Buoyancy

$$F_{\text{UBii,bz}} = F_{\text{rel}} \times (1/32 \cdot B^2 R_H^4 \Omega_H^2 / c / E_{\text{LEP}}) \times Q_{\text{wave}} \times (ac/(2R_H)) \times (1+t_{\text{var}})$$

$R_H = GM/c^2$ (horizon radius)
 $\Omega_H = ac/(2R_H)$ (BH angular velocity)
 Source: BB_C_Equations item 1147

$$F_{\text{UBii,bz2}} = F_{\text{rel}} \times (?(16p) \cdot F_{\text{BH}}^2 \cdot \Omega_{\text{BH}}^2 / c / E_{\text{LEP}}) \times Q_{\text{wave}} \times (ac^3/(2GM)) \times 0.05^{-1}$$

$F_{\text{BH}} = B \cdot p \cdot r_H^2$ (BH magnetic flux thread; EHT-calibrated)
 Source: BB_C_Equations item 967, 1316

2.6 Arnett Supernova Buoyancy

$$F_{\text{UBii,arnett}} = F_{\text{rel}} \times (M_{\text{Ni}} \cdot e_{\text{Ni}} / t_d / E_{\text{LEP}}) \times Q_{\text{wave}} \times e^{-t/t}$$

M_{Ni} = nickel mass ($\sim 0.5 M_{\odot}$)
 e_{Ni} = nickel decay energy
 $t_d^2 = 3M/(4pcv^2)$ (diffusion time)
 Source: BB_C_Equations item 1035

2.7 TOV Hydrostatic Equilibrium Buoyancy

$$F_{UBii,tov} = -F_{rel} \times (dP/dr = -Gm(r)/r^2 \cdot (1 + P(r)/(c^2)) \times (1 + 4\pi r^3 P(r)/(m(r)c^2)) / (1 - 2Gm(r)/(rc^2)) / E_{LEP}) \times Q_{wave}$$

Includes all GR corrections (Schwarzschild metric, pressure-energy coupling)
Source: BB_C_Equations item 1300

2.8 Pulsar Spin-Down Buoyancy

$$F_{UBii,puls} = -F_{rel} \times (\dot{P} = P/(2\tau)) / E_{LEP}) \times Q_{wave}$$

P = period, $\tau \sim 10^{15}$ s/s (period derivative, Vela)
 $I \cdot d\Omega/dt = -L/O$ (torque equation)
Source: BB_C_Equations item 1302

3. Stellar/Accretion Variants

3.1 Jet Velocity Buoyancy

$$F_{UBii,jetvel} = F_{rel} \times (v_j \sim v_K \cdot (r_A/r_0)^{1/2} / E_{LEP}) \times Q_{wave} \times (B/v(4\pi)) \times (1+t/t_A)$$

$v_K = v(GM/r_0)$ (Keplerian velocity at footpoint, $r_0 = 1-10$ AU)
 r_A = Alfvén radius (10–50 AU, from POETS protostellar data)
Source: BB_C_Equations item 1096, 1272

3.2 Type-I Planet Migration Buoyancy

$$F_{UBii,migr} = -F_{rel} \times (-f \cdot 2(GM_p)^2 \cdot S/(M_* \cdot a^5 \cdot (H/r)^3) / E_{LEP}) \times Q_{wave} \times [t = M_p/\dot{M}_{acc} \sim 10^6 \text{ yr}]$$

$f \sim -1.36$ (Lindblad/corotation factor, inward migration)
 S = disk surface density
Source: BB_C_Equations item 1121, 1322

3.3 Superfluid Vortex Glitch Buoyancy

$$F_{UBii,glitch} = F_{rel} \times (\dot{I} = I_s/I \cdot \Omega_0 \cdot (1 - e^{-t/t_q}) / E_{LEP}) \times Q_{wave} \times \Omega_0 \times e^{-t/t_q}$$

I_s = superfluid moment of inertia ($\sim 10^{38}$ kg·m²)
 t_q = quench timescale
 Ω_0 = angular velocity jump
Source: BB_C_Equations item 1753, 1304

4. Shock and Blast Wave Variants

4.1 J-Type Shock Buoyancy (Rankine-Hugoniot)

$$F_{UBii,jshock} = F_{rel} \times ((\rho_1 v_1^2 + P_1) / E_{LEP}) \times Q_{wave} \times (v_2/v_1) \times (\rho_2 + 1)/(\rho_1 - 1 + 2/M^2)$$

$\gamma = 5/3$ (adiabatic index)

$M = v_1/c_s$ (Mach number, from Chandra X-rays in HH 154)

Source: BB_C_Equations item 1193, 1274

4.2 Sedov-Taylor Blast Wave Buoyancy

$$F_{UBii, sedov} = F_{rel} \times ((E \cdot t^2 / \rho)^{1/5} / E_{LEP}) \times Q_{wave} \times [d/dt(Mv)=0] \times t^{2/5}$$

$E \sim 10^{51}$ erg (explosion energy)

ρ = ambient density

$R(t) = (Et^2/\rho)^{1/5}$ (self-similar blast radius)

Source: BB_C_Equations item 1207, 1288

4.3 C-Type Shock Buoyancy (Magnetized)

$$F_{UBii, cshock} = -F_{rel} \times ((\gamma B) \times B / (4p) + \rho_i \cdot \rho_{in} \cdot (v_n - v_i) / E_{LEP}) \times Q_{wave}$$

C-shocks: continuous, multi-fluid MHD (ions+neutrals+magnetic field coupled)

ρ_{in} = ion-neutral collision frequency

Source: BB_C_Equations item 1276

5. Feedback and Outflow Variants

5.1 AGN Feedback Buoyancy

$$F_{UBii, agn} = F_{rel} \times (f(v_{out}) \cdot L_{AGN}/c / E_{LEP}) \times Q_{wave} \times (\rho_{out} \cdot v_{out}) \times v_{out}^{?1}$$

Momentum-driven: $p_{term} = \rho_{out} \cdot v_{out} = f(v_{out}) \cdot L_{AGN}/c$

$\rho_{out} \sim 10\text{--}100 M_{\odot}/\text{yr}$

Source: BB_C_Equations item 1165, 1314

5.2 Angular Momentum Accretion Buoyancy

$$F_{UBii, ang} = -F_{rel} \times (\gamma \cdot r^2 \cdot \dot{M} / E_{LEP}) \times Q_{wave} \times T_B \times e^{-t/t_{disk}}$$

$T_B = B^2 r^3 / (4p)$ (magnetic braking torque)

t_{disk} = disk decay timescale

Source: BB_C_Equations item 1189, 1270

5.3 Feedback Energy Coupling Buoyancy

$$F_{UBii, coup} = F_{rel} \times ((1/2) \cdot \rho_w \cdot v_w^2 / (\rho_{acc} \cdot c^2 \cdot 10) / E_{LEP}) \times Q_{wave} \times 0.05\text{--}0.1$$

$e_f = E_{kin} / (\rho_{acc} \cdot c^2) \sim 0.05\text{--}0.1$ (coupling fraction)

Source: BB_C_Equations item 1331, 1554

6. Structure Formation Variants

6.1 Press-Schechter Halo Mass Function Buoyancy

$$F_{UBii,ps} = F_{rel} \times (dn/dM = v(2/p) \cdot (\theta_0/M) \cdot (d_c/s(M)) \cdot |d \ln s/d \ln M| \cdot \exp(-d_c^2/(2s^2))) / E_{LEP} \\ \times Q_{wave} \times \theta_0 \text{ (Gaussian part from } s \sim d_c)$$

$d_c \sim 1.69$ (critical collapse overdensity)
Source: BB_C_Equations item 877, 1574

6.2 Structure Growth Rate Buoyancy

$$F_{UBii,grow} = -F_{rel} \times (d'' + 2H \cdot d' = (3/2) \cdot \Omega_m \cdot H^2 \cdot d/a^3 / E_{LEP}) \times Q_{wave}$$

$D(a) = \Omega_m/2 \cdot \int_0^a da' / (a' H(a')/H_0)^3$ (growth factor)
Growing mode: $D \propto a$ in matter era, suppressed by dark energy
Source: BB_C_Equations item 1371, 1244

7. GRB Variants

7.1 GRB Fireball Expansion Buoyancy

$$F_{UBii,fire} = F_{rel} \times (G(r) = r/R_0 \text{ (} r < R_s), G=1 \text{ (} r > R_s) / E_{LEP}) \times Q_{wave}$$

$R_0 \sim 10^7$ cm (initial fireball radius)
 $R_s = \gamma^2 R_0$ (saturation radius)
Source: BB_C_Equations item 1482, 1306

7.2 GRB Afterglow Synchrotron Buoyancy

$$F_{UBii,after} = -F_{rel} \times (F_{\nu} \propto \nu^{-(p-1)/2} \cdot t^{-3(p-1)/4} \text{ (} \Omega_m < \Omega_c) / E_{LEP}) \times Q_{wave}$$

$p \sim 2.2-2.5$ (electron power-law index)
Electrons accelerated by DSA, spectrum $N(E) \propto E^{-p}$
Source: BB_C_Equations item 1227, 1308

8. References

- grok_share_7514fe.txt lines 2443-2680 (BB_C_Equations_04Sept2025.pdf Part 1 catalogue)
- PAPER_196: Triadic Master Equation System
- PAPER_197: F_U_Bi_i Extended Integral